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Google Scholar: https://scholar.google.com/citations?hl=en&user=v7X_OhYAAAAJ
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RESEARCH INTERESTS

My primary research interests lie in group theory and representation theory. I particularly focus on the structural theory of Chevalley and twisted Chevalley groups, while also having a broader interest in algebraic groups, Lie groups, and Lie algebras.

EDUCATION

Doctor of Philosophy - Ph.D. (Mathematics) Indian Institute of Technology Bombay Thesis Title: Some Properties of Twisted Chevalley Groups Thesis Advisor: Prof. Shripad M. Garge	January 2020 - May 2025 Mumbai, Maharashtra, India
Master of Science - M.Sc. (Mathematics) Indian Institute of Technology Bombay	July 2017 - May 2019 Mumbai, Maharashtra, India
Bachelor of Science - B.Sc. (Mathematics) The Maharaja Sayajirao University of Baroda	June 2014 - April 2017 Vadodara, Gujarat, India

CURRENT POSITIONS

Postdoctoral Fellow Department of Mathematics, Bar-Ilan University	1st October 2025 - present Ramat-Gan, Israel
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PAST POSITIONS

Research Associate Department of Mathematics, IIT Bombay	8th January 2025 - 28th May 2025 Mumbai, Maharashtra, India
Visiting Researcher Harish-Chandra Research Institute (HRI)	July 2025 - September 2025 Prayagraj, Uttar Pradesh, India

PUBLICATIONS AND PREPRINTS

- (1) Shripad M. Garge and Deep H. Makadiya, *On Normal Subgroups of Twisted Chevalley Groups over Commutative Rings*, Accepted in Journal of Algebra (2025). ArXiv link: [arXiv.2502.04766](https://arxiv.org/abs/2502.04766).

- (2) Shripad M. Garge and Deep H. Makadiya, *Normalizer of Twisted Chevalley Groups over Commutative Rings* Accepted in Journal of Pure and Applied Algebra (2025). ArXiv link: [arXiv:2503.16894](https://arxiv.org/abs/2503.16894).
- (3) Shripad M. Garge and Deep H. Makadiya, *Triangular and Unitriangular Factorization of Twisted Chevalley Groups*, Accepted in Israel Journal of Mathematics (2025). ArXiv link: [arXiv.2505.20224](https://arxiv.org/abs/2505.20224).

TEACHING EXPERIENCE

Teaching Assistantship at IIT Bombay

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| (1) MA 111 : Calculus 2 | Autumn 2020-21 (Second Half) |
| Taught by Prof. Rekha Santhanam and Prof. Debanjana Mitra | |
| (2) MA 106 : Linear Algebra | Spring 2020-21 (First Half) |
| Taught by Prof. Shudhir Ghorpade and Prof. Dipendra Prasad | |
| (3) MA 108 : Differential Equations | Spring 2020-21 (Second Half) |
| Taught by Prof. Prachi Mahajan and Prof. U. K. Anandavardhanan | |
| (4) MA 109 : Calculus 1 | Autumn 2021-22 (First Half) |
| Taught by Prof. Sourav Pal and Prof. Mayukh Mukherjee | |
| (5) MA 111 : Calculus 2 | Autumn 2021-22 (Second Half) |
| Taught by Prof. Bata K Das and Prof. Saurav Bhaumik | |
| (6) MA 106 : Linear Algebra | Spring 2021-22 (First Half) |
| Taught by Prof. G K Srinivasan and Prof. S Sivasubramanian | |
| (7) MA 108 : Differential Equations | Spring 2021-22 (Second Half) |
| Taught by Prof. Suresh Kumar and Prof. Santanu Dey | |
| (8) MA 419 : Basic Algebra | Autumn 2022-23 (Full Semester) |
| Taught by Prof. Jugal K Verma | |
| (9) MA 414 : Algebra 1 | Spring 2022-23 (Full Semester) |
| Taught by Prof. Ronnie M Sebastian | |
| (10) MA 419 : Basic Algebra | Autumn 2023-24 (Full Semester) |
| Taught by Prof. Tony J Puthenpurakal | |

Pooja Coaching Institute, Vadodara

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| (1) Mathematics Tutor for IIT JAM and CSIR-UGC NET/JRF Aspirants | July 2019 - January 2020 |
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CONFERENCES, WORKSHOPS AND OTHER TALKS

Conferences

- (1) Conference on **“Groups and Representations”** held at IIT Bombay, Mumbai – 2024
Presented my ongoing work with Shripad M. Garge on classifications of the automorphisms of twisted Chevalley groups over commutative rings.
- (2) Conference on **“Representations of Groups and Algebras”** held at IMSc, Chennai – 2025
Give a poster presentation on classifying the normal subgroups of twisted Chevalley groups over commutative rings.

Workshop Participations

- (1) NCM Workshop on **“Lie Groups and Representation Theory”** held at IISER, Berhampur – 2024
- (2) National Workshop on **“Problem Solving Technique in Mathematics”** (Supported by UGC) at Faculty of Science, M.S. University of Baroda, Vadodara – 2017
- (3) Workshop on **“Problem Solving & Enrichment in Mathematics”** held at St. Xavier’s College, Ahmadabad – 2016

Seminar Series at IIT Bombay

- (1) Seminar based on the book **“Linear Algebraic Groups”** by T. A. Springer – 2023 to 2024

- (2) Seminar based on Steinberg's famous paper "**Regular elements of semi-simple algebraic groups**" – 2023
- (3) Seminar based on the book "**Nilpotent Orbits of Lie algebras**" by D. H. Collingwood and W. M. McGovern – 2024
- (4) Seminar based on the book "**Spherical Functions on a Group of p -adic Type**" by I. G. Macdonald – 2025

Student Seminars at IIT Bombay

- (1) Give two seminars on "**Generator Relation Presentation of the Group $SL_n(F)$ and Steinberg groups**", student seminar at IIT Bombay – May 2024
- (2) Give seminar on "**Automorphism of Local Rings which Fixes the Maximal Ideal**", student seminar at IIT Bombay – February 2024
- (3) Give seminar on "**Representation Theory of $SL_2(\mathbb{C})$** ", student seminar at IIT Bombay – March 2023

ACADEMIC ACHIEVEMENTS

- Secured **AIR 10 (out of 10834)** in **IIT-JAM** Examination for M.Sc. Mathematics – 2017
- Clear **NET/JRF** examination with All India **JRF Rank 53** (with the first attempt during the first semester of M.Sc.) – 2017
- Clear **NET/JRF** examination (second time) with All India **JRF Rank 67** – 2018
- Clear **NET/JRF** examination (third time) with All India **JRF Rank 19** – 2019
- Qualify **GATE** examination in the first attempt – 2019
- Secured **Rank 1** in **MS University** Entrance Examination for M.Sc. Mathematics – 2017
- Secured **Rank 4 (Gujarat State)** in Madhava Mathematics Competition (MMC) – 2017

SKILLS

- **Languages:** Proficient in reading, writing, and speaking in English, Hindi and Gujarati
- **Coding:** LaTeX, sagemath, C, C++

REFERENCES

- (1) Prof. Shripad M. Garge
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IIT Bombay, Mumbai, India
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- (2) Prof. Dipendra Prasad
Emeritus Fellow
Department of Mathematics
IIT Bombay, Mumbai, India
E-mail: prasad.dipendra@gmail.com, dprasad@math.iitb.ac.in
- (3) Prof. Maneesh Thakur
Professor
ISI Bangalore, India
E-mail: maneesh@isibang.ac.in
- (4) Prof. Saurav Bhaumik
Assistant Professor
Department of Mathematics
IIT Bombay, Mumbai, India
E-mail: bhaumiksaurav@gmail.com, saurav@math.iitb.ac.in

DECLARATION

I hereby declare that all the above-mentioned information provided by me stands true to the best of my knowledge and if there is any discrepancy found, I shall be liable for any disciplinary action deemed suitable by the competent authority.